

Apostol, Chapter 1 - Exercise 15

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Prove that every $n \geq 12$ is the sum of two composite numbers.

Let n be even such that $n \geq 8$. Then $n - 4$ is an even number greatest or equal than 4, thus a composite number. Therefore, for every even $n \geq 8$ we can write it as the sum of two composite numbers as follows,

$$n = 4 + (n - 4).$$

Now let n be odd such that $n \geq 13$. Then $n - 9$ is an even number greatest or equal to 4, thus a composite number. Therefore, for every odd $n \geq 13$ we can write it as the sum of two composite numbers as follows,

$$n = 9 + (n - 9).$$

That is, every $n \geq 12$ can be written as the sum of two composite numbers.